Anna Ochab-Marcinek, Ph.D., D.Sc.

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Specialist in energy systems analysis and computational methods | National Centre for Energy Analysis (PSE)

Skills

- **Python**, **C++**, **Linux** shell scripting (Bash), **Machine Learning**, optimization, image analysis, unit testing, Git, Jira, Jupyter, Docker, **Gurobi**.
- mySQL, HTML, CSS, JavaScript, PHP, C, Fortran, Maple, Mathematica. I have been exposed to: R, Arduino C++ programming, Matlab, Excel VBA.
- With a background in statistical physics, I have spent most of my career involved in modelling using the theory of stochastic processes. I have a solid knowledge of statistics.
- I have strong analytical skills.

Projects

ZefirLib for Local Governments | *Python (Gurobi), Bash, Excel VBA* Nov 2025 – Present

- I analysed the code of ZefirLib software packages, authored parts of technical documentation to address existing gaps, and coordinated team contributions to user-friendly documentation version.
- I built experience in Zefir model input preparation and results analysis, automated preprocessing, execution, and postprocessing workflows.

Statistical analysis of Raman hyperspectral imagery | *Python (PyTorch, scikit-learn); data preprocessing, Principal Component Analysis, NFINDR unmixing, Kullback-Leibler divergence* May 2025 – Oct 2025

- I developed Machine Learning models for Raman spectra classification.
- I statistically analysed Raman hyperspectral imagery of infected pseudo-tissues in order to identify the hallmarks of bacteria.

Automated detection and analysis of bacterial clusters in microscopic images | *Python (PyTorch, TensorFlow, Scipy, Skimage, Numpy, Pandas, PyTest), C++* Sep 2023 – Oct 2025

- I developed the software for automated detection and analysis of numbers, sizes, orientations of individual bacteria within bacterial clusters in microscopic 3-D images using the libraries for optimization (SciPy) and image analysis (Skimage).
- I developed the methods for deconvolution of the point spread function (PSF) from the microscopic 3-D image of a known probe. To find the PSF, I applied non-traditional Machine Learning (optimization by differentiable rendering) with PyTorch and Tensorflow.

Simulation of gene expression evolution in a fitness landscape as Levy Flights: fitnesslandscapestudent | *C++, Python, Jupyter* Oct 2022 – May 2023

- A research project for stochastic modelling of gene expression evolution (aom branch is my initial individual work, master branch is the further collaboration with my student).
- Uses my C++ interface for 2D bilinear interpolation with GSL, described in the next point.

C++ interface for 2D bilinear interpolation with GNU Scientific Library (GSL): interpolation2dcpp | *C++, C* May 2022 – Jun 2022

- I created this object-oriented C++ interface because the original GSL routines are in C.

Statistical analyses of epidemiological data: govpl | *Python, Jupyter* Jun 2020 – Dec 2022

- My early epidemiological analyses (see next point) attracted enough attention that I was invited to participate in the COVID-19 Interdisciplinary Advisory Team to the President of the Polish Academy of Sciences.
- I performed statistical analysis of public epidemiological data.
- I co-authored 27 position papers, which were widely picked up by the media.
- Because of my work within the Team, the Polish Press Agency published an interview with me about the caveats of interpreting epidemiological models: "Dr. Ochab-Marcinek: it is difficult to predict the course of epidemics using computer simulations" (in Polish).

Epidemiological data analysis during the early stages of the COVID-19 pandemic: Mar 2020 – Jun 2020
COVID-19-MZ_GOV_PL | *Python, Jupyter*

- I identified a need in Polish society for a source of more comprehensive, explained, and visualised information on COVID-19 epidemiology.
- At that time, the Ministry of Health did not publish data in a human-readable format, only bitmaps with numbers on Twitter. Although Rogalski's spreadsheet project appeared, it lacked some data categories and graphs types.
- I used web scraping, image filtering of decorative backgrounds, and OCR to collect the missing data from the Ministry of Health's Twitter.
- I calculated various statistics and published the graphs with my comments on the IPC PAS website (archived) and on Facebook.
- The project was unique in that it included plots and analytical comments that were not available elsewhere. These analyses attracted sufficient attention for me to be invited to participate in the COVID-19 Interdisciplinary Advisory Team under the President of the Polish Academy of Sciences.

Various programming projects for research | *C++, Python* 2002 - 2023

- Stochastic gene expression modelling, stochastic cancer growth modelling, stochastic cell population growth modelling.

Experience

Coordinator Specialist *National Centre for Energy Analysis (NCAE), Polskie Sieci Elektroenergetyczne S. A.* Nov 2025 – Present
Researcher *Dioscuri Centre for Physics and Chemistry of Bacteria, Institute of Physical Chemistry, Polish Academy of Sciences* Jan 2024 – Oct 2025
Member of the COVID-19 Interdisciplinary Advisory Team to the President of the Polish Academy of Sciences Jun 2020 – Dec 2022
Principal Investigator *Biophysical Chemistry Group, Institute of Physical Chemistry PAS* Jan 2013 – Dec 2023
Postdoctoral researcher *Institute of Physical Chemistry, PAS* Apr 2009 – Dec 2012
Postdoctoral researcher *University of Augsburg, Germany* Oct 2007 – Sep 2008
Assistant Professor *Jagiellonian University, Krakow, Poland* Oct 2006 – Sep 2009

Publication record

- 26 peer-reviewed research papers recorded in Web of Science data base. H-index: 16; 764 times cited (WoS 2025).

Education

Institute of Physical Chemistry, Polish Academy of Sciences – Habilitation (DSc) in Chemistry Oct 2018
Jagiellonian University – Ph.D. in Physics Sep 2006
Jagiellonian University – M.Sc. in Physics Jun 2002

Ability to cooperate in a project team

- I pay a lot of attention to code quality, including proper documentation and commenting so that the code is easy for others to understand. As a C++ backgrounder, I favour explicit typing and the use of static type checking in Python.
- I proactively identify technical problems that hinder teamwork and propose solutions.
- I author Confluence guides on software troubleshooting, promoting team knowledge sharing.
- In the past I have been both a member and leader of research groups: I have been principal investigator on 4 grants and contractor on 4 grants. I have also received 1 travel grant for a collaborative research visit abroad. I have supervised students (PhD, MSc, internships), including the supervision of programming projects.

Languages

- English: Spoken and written on a daily basis at work in the multi-national environment of NCAE at PSE. 2002: Cambridge Certificate in Advanced English.
- German (intermediate)
- Russian (intermediate)
- Polish (native)

GDPR clause

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